Day 11 Activity: Wrangling the `penguins`

Tuesday, June 2, 2026

Your Name Here

Setup

```r
library(tidyverse)
library(palmerpenguins)
```

Today's activity uses the `penguins` dataset again. Glimpse it before you start:

```r
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

You now have all six verbs: yesterday's `group_by()` and `summarise()`, and today's `filter()`, `arrange()`, `select()`, and `mutate()`.

Part 1 — Match the verb

Each task can be done with **one** verb (one task needs two). Name the verb(s) you'd use, no code yet.

Table 1: Fill in the third column.

#	Task	Verb(s)
1	Keep only the Gentoo penguins	
2	Add a column of body mass in kilograms	
3	Sort the penguins heaviest-first	
4	Keep only the `species`, `island`, and `body_mass_g` columns	
5	One row per species, each with its average body mass	
6	Drop the penguins whose `sex` is missing	

Part 2 — filter()

Write a **single pipeline** for each. Run each one and glance at the number of rows.

2a. Penguins heavier than 5000 g.

```
# Your code:
```

2b. Adelie penguins that live on Dream island (both conditions).

```
# Your code:
```

2c. Penguins that are Adelie *or* Gentoo. Write it **two ways**, once with `|` and once with `%in%`.

```
# Your code:
```

2d. Debug it. This errors. What's the message, why does it happen, and what's the fix?

```
penguins |>
  filter(species = "Chinstrap")
```

[Your explanation and fix.]

Part 3 — mutate() and arrange()

3a. Add a column `bill_ratio` = bill length ÷ bill depth. Then `arrange()` so the largest ratio is on top. (Which species ends up at the top?)

```
# Your code:
```

[Which species is on top?]

3b. Add a column `size` that is `"large"` when `body_mass_g > 4000` and `"small"` otherwise. (Hint: `ifelse()`.)

```
# Your code:
```

3c. *Reduction or transformation?* You want the **single heaviest** body mass in the whole dataset. Does that belong in `mutate()` or `summarise()`? Write it, and say why.

```
# Your code:
```

[Why that verb?]

Part 4 — select()

4a. Keep only `species`, `sex`, and the two bill columns.

```
# Your code:
```

4b. Do the same selection of the two bill columns using a **helper** (`starts_with()` or `ends_with()`) instead of naming them.

```
# Your code:
```

4c. Select `body_mass_g` but rename it to `mass_g` in the same step.

```
# Your code:
```

Part 5 — Put it together

Answer this in **one pipeline**: *Among penguins with a known sex, what is the average flipper length for each species/sex combination, and how many penguins is each average based on? Show the longest-flippered group first.*

Before you run it, **predict** how many rows the result has.

Prediction (rows): […]

```
# Your pipeline (filter -> group_by -> summarise -> arrange):
```

Now turn the same filtered data into a plot of `body_mass_g` by `species`, filled by `sex`:

```
# Your plot (remember: switch |> to + when you reach ggplot):
```

One or two sentences: what does the table or plot show?

[Your answer.]

Part 6 — Restyle it

The pipeline below works, but it's unreadable. Rewrite it following the style guide: a space before each `|>`, `|>` at the end of the line, each named argument of `summarise()` on its own line indented two spaces, and the closing `)` on its own line.

```
penguins|>filter(!is.na(sex))|>group_by(species,sex)|>summarize(n=n(),avg_mass=mean(body_mas
```

```
# Your restyled version:
```

> **Tip**
>
> Once you've done it by hand, try the `styler` package: `Cmd/Ctrl + Shift + P`, then type "styler" and pick *"Style active file."* Did it match yours?

Wrap-up

One-sentence reflection: of the four verbs today, which felt most natural and which will you have to look up again next week?

[Write one sentence.]

Code appendix

```
library(tidyverse)
library(palmerpenguins)
glimpse(penguins)
# Your code:

# Your code:

# Your code:

penguins |>
  filter(species = "Chinstrap")
# Your code:

# Your code:

# Your code:

# Your code:

# Your code:

# Your code:

# Your pipeline (filter -> group_by -> summarise -> arrange):

# Your plot (remember: switch |> to + when you reach ggplot):

penguins|>filter(!is.na(sex))|>group_by(species,sex)|>summarize(n=n(),avg_mass=mean(body_mas
# Your restyled version:
```